CS5290 Memory-Centric Computer Architecture · Results Report (24 October 2026)

Protecting the Right Weights on a Noisy ReRAM Chip

Phase 1 results for BERT-Base and DistilBERT on a hybrid SLC/MLC crossbar

Main findings

1. Storing a transformer only in dense two-bit cells (MLC) does not work: at realistic programming noise, BERT-Base falls from 90% to near coin-flip accuracy.
2. Keeping a small share of SVD components in reliable one-bit cells (SLC) recovers almost all of this loss, but only if the right components are chosen.
3. Ranking components by their **singular value (magnitude)** is as good as the best placement we could construct. A Fisher-information estimate of each component's expected damage on the chip reaches the same accuracy (within one sentence in 872 on average over 600 chips), while a HyFlexPIM-style gradient loses 3–12 points at high noise.
4. With magnitude ranking, 5–10% SLC suffices at 5% noise and about 30% at 7.5% noise, using roughly 60–70% of the memory cells of an all-SLC chip.
5. The conclusions hold on a more realistic chip (read noise, level-dependent noise, fixed ADC, IR drop), and a 6-bit ADC is enough for a hybrid chip.
6. Stuck cells are the one problem placement cannot solve: all-MLC fails at 0.1% stuck cells and all-SLC at 1%. This motivates fault-aware mapping, our next step.

1 Problem and Questions

A ReRAM crossbar computes a matrix–vector product inside memory: each weight is stored as the conductance of resistive cells, and the currents of a column add up to the result. This avoids most of the data movement of conventional hardware, but a cell cannot be programmed exactly, so every weight carries random error. A cell can hold one bit (SLC), with two well-separated levels that tolerate noise, or two bits (MLC), which is denser but more fragile. In our set-up an 8-bit weight needs 7 SLC or 4 MLC cells per sign.

HyFlexPIM (ISCA 2025) splits each weight matrix with singular value decomposition (SVD) into components, stores the most important ones in SLC and the rest in MLC, and ranks importance by the loss gradient. We asked: (i) does this hybrid scheme hold up in a bit-accurate hardware simulation, (ii) which components should be protected, and how many, and (iii) which hardware effects still break it.

2 Method

We used two pretrained sentiment classifiers, **BERT-Base** (72 weight matrices on the crossbar) and **DistilBERT** (36), evaluated on all 872 validation sentences of SST-2. Each matrix is factorised with SVD (384 components for a 768×768 layer) and the model is fine-tuned for 8 epochs. The simulator stores 8-bit weights on differential cell pairs split into bit slices. Programming noise σ is added to every cell as a fraction of the maximum conductance. One *chip* is one random draw of all cell errors; results are averaged over 5–20 chips, and we also report the worst chip. Every placement is evaluated on the same chips, so differences between placements are not due to chance draws. The SVD model is built once: a placement only decides which cells each component is stored in.

2.1 Placements

- **All SLC** and **All MLC**: the two extremes, giving the best and the cheapest chip.
- **Random**: a share of components chosen at random, as a control.
- **Magnitude**: the components with the largest singular values σ_r.
- **Gradient (HyFlexPIM-style)**: $\sum_b |\partial L_b / \partial \sigma_r|$, absolute loss gradients summed over batches of 32 training sentences with dropout on.
- **Fisher**: the expected rise in loss if component r is left in MLC,

$$D_r = \sum_{e \in \text{entries of } r} F_e \, \mathrm{Var}(\text{noise}_e),$$

summed over every stored entry of the component (its row $\sigma_r v_r$ and its column u_r). F_e is the empirical Fisher information of entry e over 2,048 training sentences, and the noise variance follows our cell model: fixed cell noise becomes weight noise in proportion to the 8-bit scale of the crossbar row that holds the entry.

Two design choices follow from how the chip behaves. The gradient takes absolute values per batch, because on a trained model the gradients of individual sentences point in opposite directions and would largely cancel in a single sum. The Fisher score covers every stored entry of a component, because chip noise disturbs all of them, not only σ_r.

3 Baseline: the Cost of SVD

Model	Original FP32	Original INT8	SVD, before fine-tune	SVD, after fine-tune
BERT-Base	92.43	91.86	89.33	90.14
DistilBERT	91.06	90.37	89.68	89.11

Table 1: SST-2 accuracy (%) with no hardware error.

8-bit quantisation costs only about 0.6 points. SVD costs about 2 points even after fine-tuning; for DistilBERT, 8 epochs slightly over-fitted (HyFlexPIM uses 1–3). This loss is paid before any hardware error and applies equally to every placement. With no noise, the simulated chip matches the SVD model to within 0.5 points for every placement, confirming that the simulator adds no hidden error.

4 Programming Noise

Placement	$\sigma = 0$	2%	3%	5%	7.5%	10%
All SLC	89.9	89.3	89.1	87.8	85.9	79.7
All MLC	90.0	87.3	83.9	66.3	52.2	51.3
Magnitude, 20% SLC	89.7	89.9	**90.0**	88.6	83.5	72.0
Fisher, 20% SLC	89.1	90.1	**90.0**	**89.0**	**84.5**	**73.7**
Gradient, 20% SLC	89.8	**90.2**	**90.0**	88.0	79.3	67.7
Random, 20% SLC	90.0	89.9	87.7	78.8	58.8	49.1

Table 2: BERT-Base, mean accuracy (%) over 5 chips. DistilBERT behaves the same way (Appendix A).

All-MLC collapses once noise reaches 5%. Protecting 20% of the components by magnitude or Fisher recovers nearly all of the loss, the gradient recovers less at high noise, and a random 20% recovers little: *which* components are protected matters more than how many.

5 Choosing and Sizing the Protected Set

Noise	Placement	0%	5%	10%	20%	30%	50%	100%
$\sigma = 5\%$	Magnitude	66.8	**88.0**	**88.6**	88.7	**88.9**	**88.9**	88.0
	Fisher		87.9	**88.6**	**88.8**	**88.9**	**88.9**	
	Gradient		84.2	88.2	88.4	88.4	88.1	
	Random		72.5	73.6	76.2	77.8	83.4	
$\sigma = 7.5\%$	Magnitude	53.0	79.4	81.8	84.1	**85.5**	85.6	85.7
	Fisher		**80.0**	**82.2**	**84.2**	85.1	**85.8**	
	Gradient		67.5	79.0	81.2	82.4	83.2	
	Random		53.6	54.2	55.2	57.5	67.3	

Table 3: BERT-Base, mean accuracy (%) over 20 chips by share of components in SLC. DistilBERT, $\sigma = 3\%$ and worst-chip results are in Appendix A.

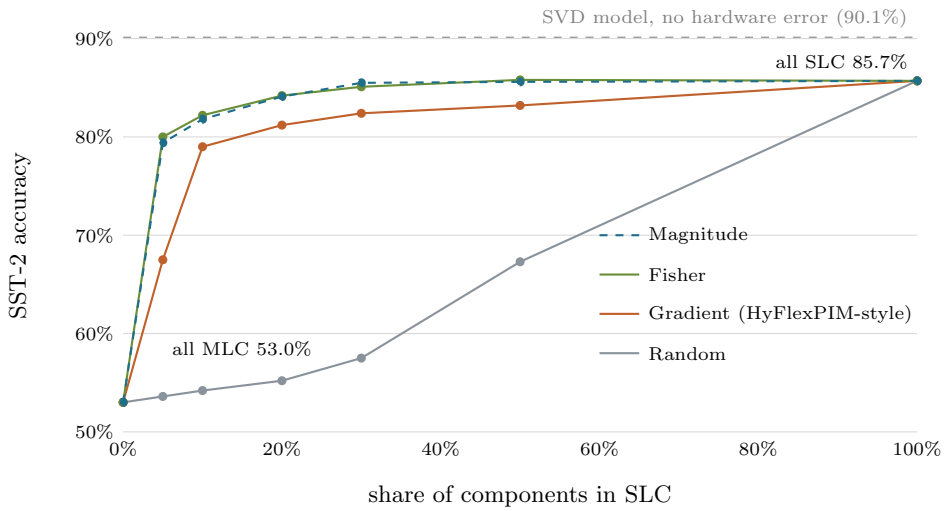


Figure 1: BERT-Base at $\sigma = 7.5\%$, mean of 20 chips. 0% and 100% are all-MLC and all-SLC. Magnitude (dashed) and Fisher lie on top of each other.

5.1 Magnitude and Fisher are equivalent; the gradient falls behind

Because every placement is tested on the same 20 chips, we can compare them chip by chip. Over all 300 chips of the SLC-share study (3 noise levels, 5 SLC shares), Fisher differs from magnitude by $+0.4 \pm 0.7$ sentences per chip on BERT-Base and -0.6 ± 0.7 on DistilBERT (mean $\pm$ standard error, out of 872): no measurable difference. The gradient is clearly worse, by 17.3 ± 2.0 and 7.8 ± 1.1 sentences; at $\sigma = 7.5\%$ it trails by 3 points with 20% SLC and by 12 points with 5% SLC. Random selection is far behind everywhere.

5.2 Why magnitude is enough

After SVD, component r is stored as the row $\sigma_r v_r$ and the column u_r. Each row is quantised with its own scale, set by its largest value, so the same cell noise produces a weight error proportional to σ_r. The harm a component does when left in MLC is its sensitivity times the size of the noise it receives, and the Fisher score estimates exactly this product over all of the component's stored entries. That this principled estimate ends up choosing essentially the same components as magnitude shows that noise size, set by σ_r, dominates the damage: magnitude ranking is close to the best placement available under this noise model. It also needs no data or gradients. The gradient falls short because it measures sensitivity to small changes of σ_r alone, ignoring both the size of the noise and the other stored entries.

5.3 How much SLC is needed

With magnitude ranking, the SLC share needed to come within half a point of an all-SLC chip is about 5% at $\sigma = 3\%$, 5–10% at $\sigma = 5\%$ and about 30% at $\sigma = 7.5\%$; DistilBERT needs 30–50% at $\sigma = 7.5\%$. Counting only weight cells, a share f in SLC needs $7f + 4(1 - f)$ cells per weight against 7 for all-SLC: 61% of the cells at 10% SLC and 70% at 30% (all-MLC: 57%). At moderate noise a hybrid chip therefore approaches MLC density at SLC accuracy.

6 Stuck-At Faults

A stuck cell reads fully on or fully off, whatever was written. We made a share of all cells stuck, half on and half off, with no programming noise and 20% SLC.

Faults are far more harmful than noise, because a stuck cell in a high-order slice shifts its weight by a large, fixed amount. All-MLC fails when one cell in a thousand is stuck, and even all-SLC fails at 1%. Up to 0.2%, magnitude and Fisher with 20% SLC are as good as all-SLC; the gradient degrades sooner. Near the collapse point, accuracy varies by 6–9 points from chip to chip, so survival depends on *where* the faults land. Reported stuck-cell rates for real arrays are of the order of a few percent, well past the point where every placement fails, so fault-aware mapping is required.

Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
All SLC	89.9	88.3	85.6	**79.0**	**62.7**	51.4	49.6
All MLC	90.0	58.6	52.6	50.8	51.2	49.6	49.6
Magnitude, 20% SLC	89.7	89.6	**88.6**	76.8	57.9	51.0	50.0
Fisher, 20% SLC	89.1	**89.7**	88.4	75.6	55.5	49.5	50.3
Gradient, 20% SLC	89.8	89.4	87.5	71.0	52.8	50.2	49.5
Random, 20% SLC	90.0	67.9	55.8	50.7	50.6	50.2	49.9

Table 4: BERT-Base, mean accuracy (%) over 20 chips versus the share of stuck cells.

7 A More Realistic Chip

We added the main effects missing from the basic simulator and tested each on its own, after checking the new simulator against exact results.

Condition	All SLC	All MLC	Magnitude	Fisher	Gradient
No non-idealities	89.7	90.4	90.6	89.6	90.3
Read noise 2% (fresh at every read)	89.0	77.3	**90.1**	89.7	89.8
Read noise 5%	80.0	53.0	82.5	**82.7**	76.6
Middle MLC levels 2× noisier ($\sigma = 5\%$)	88.2	62.5	**88.4**	**88.4**	87.8
1% stuck cells, scattered	**61.3**	50.7	56.4	55.6	52.6
1% stuck cells, 3×3 clusters	**66.7**	50.8	54.0	53.1	51.5
1% stuck cells, whole bit lines	**57.0**	50.4	50.4	52.2	51.0
Fixed ADC with bit-serial inputs, 6 bits	89.3	83.0	90.0	**90.5**	89.6
Fixed ADC with bit-serial inputs, 5 bits	51.1	51.1	51.1	51.6	53.0
IR drop, $r = 3 \times 10^{-4}$	88.9	89.8	90.4	**91.1**	90.5
IR drop, $r = 3 \times 10^{-3}$	58.5	**79.1**	76.8	72.8	76.4

Table 5: BERT-Base, mean accuracy (%); hybrids keep 20% of components in SLC. Digital SVD model: 90.1%. r is the resistance of one wire segment relative to a cell's on-resistance. DistilBERT results are in Appendix A.

- **Read noise and level-dependent noise** behave like programming noise: all-MLC fails, while magnitude and Fisher stay within about a point of all-SLC. The gradient loses 6 points at 5% read noise.
- **ADC resolution.** With the ADC range fixed by calibration, 6 bits is enough for all-SLC and the hybrids, all-MLC needs 7 bits, and 5 bits fails for every placement. With 5% programming noise (DistilBERT), 6 bits still holds the magnitude hybrid at 88.5% while all-MLC drops to 56.0%. Since ADCs dominate crossbar energy, this sets the target for the hardware-cost study.
- **IR drop** has no effect at realistic wire resistance ($r \lesssim 3 \times 10^{-4}$ in 128×128 tiles). When large, it hurts SLC most: an SLC cell storing a 1 sits at full conductance and loses the most current to wire resistance.
- **Fault patterns.** Clustered faults affect fewer weights than scattered ones, and whole broken bit lines are worst. At 1% every placement is already near failure, confirming Section 6.

Random selection was not repeated for these effects: as a control, it is already far behind in every experiment above.

8 Relation to HyFlexPIM

Our results agree with HyFlexPIM that a hybrid SLC/MLC layout works, but not on how to rank components. HyFlexPIM reports that its gradient ranking beats ranking by singular value by about a point or less, on MRPC and CoLA at a single noise level. We find no difference at mild noise ($\sigma = 3\%$) and the reverse at high noise, so the two findings need not conflict. Three differences in set-up matter:

- **Noise model.** HyFlexPIM applies relative noise to whole weights, $\tilde{W} = W \odot (1 + \eta)$, so small weights are almost exact. We add noise to each cell, as real hardware and standard simulators do, so even a zero weight carries the error of its two drifting cells. The paper gives no reason for its choice; its citations support only the noise level, a 4.04% bit error rate measured on 3 million real MLC cells, which is itself a per-cell quantity.

- **Noise range.** The paper evaluates one fixed noise level. Our sweeps cover 2–10%, and the ranking differences appear only at $\sigma \geq 5\%$.
- **When gradients are taken.** HyFlexPIM records gradients during fine-tuning; ours are taken on the trained model. A re-run of their fine-tuning was not pursued, since magnitude already matches the best placement we could construct.

9 Conclusions

- A hybrid SLC/MLC layout is necessary: all-MLC storage fails at realistic noise.
- Ranking components by singular value is the recommended placement. A Fisher estimate of each component's expected damage chooses essentially the same components and reaches the same accuracy, which shows that magnitude is close to the best placement under this noise model; it needs no data or gradients. A HyFlexPIM-style gradient is clearly weaker at high noise.
- The SLC share should follow the expected noise: about 10% at moderate noise and 30% at high noise, saving 30–40% of the cells of an all-SLC chip.
- The conclusions hold under read noise, level-dependent noise, a fixed 6-bit ADC and realistic IR drop.
- Stuck cells break every placement at rates below those reported for real arrays, so fault-aware mapping is necessary, not optional.

10 Limitations

- Each non-ideality was tested on its own; their combined effect is not yet measured. The IR-drop model is static, and the ADC range is set by the maximum calibration value rather than a percentile.
- Inputs are quantised to 8 bits with one scale per batch of sentences, so accuracy shifts slightly with batch size (by up to about one point without noise). All placements within an experiment use the same batches, so comparisons are unaffected; per-sentence input scaling will be added.
- The SVD step costs about 2 points, and DistilBERT's fine-tuning needs early stopping.
- All results are on SST-2; MRPC and the CNN models follow in Phase 2. Literature figures for stuck-cell rates and wire resistance are still to be confirmed with citations.

11 Next Steps

Stuck cells are the one effect placement cannot handle. Fault-aware mapping, and its extension that jointly accounts for cell type, faults and IR drop, come next; they need no retraining, only a smarter assignment of components to crossbar positions. Noise-aware training, including a variant that applies SLC- and MLC-specific noise during fine-tuning, follows.

Models: textattack/bert-base-uncased-SST-2 and distilbert-base-uncased-finetuned-sst-2-english. Simulator: 8-bit weights, on/off ratio 100, differential cell pairs, 128×128 tiles. Ranking scores computed on 2,048 SST-2 training sentences.

A Complete Result Tables

All values are SST-2 validation accuracy (%, 872 sentences). Hybrid placements keep 20% of components in SLC unless the SLC share is varied. In each column the best hybrid placement is shown in colour; "–" marks points that were not run.

A.1 Programming noise (5 chips)

Placement	0	1%	2%	3%	5%	7.5%	10%
All SLC	89.9	89.6	89.3	89.1	87.8	85.9	79.7
All MLC	90.0	89.3	87.3	83.9	66.3	52.2	51.3
Magnitude	89.7	89.8	89.9	90.0	88.6	83.5	72.0
Fisher	89.1	89.7	90.1	**90.0**	**89.0**	**84.5**	**73.7**
Gradient	89.8	**90.4**	**90.2**	90.0	88.0	79.3	67.7
Random	**90.0**	90.1	89.9	87.7	78.8	58.8	49.1

Table 6: BERT-Base, mean accuracy versus programming noise σ.

Placement	0	1%	2%	3%	5%	7.5%	10%
All SLC	89.0	88.7	88.8	88.6	88.5	87.5	85.9
All MLC	89.1	88.4	87.9	86.1	74.5	60.4	56.0
Magnitude	89.1	88.9	88.9	88.9	88.4	**85.9**	**77.7**
Fisher	89.0	**89.0**	**89.0**	**89.1**	**88.5**	85.5	77.6
Gradient	**89.8**	88.5	88.9	88.7	87.7	83.4	69.8
Random	88.6	88.7	88.7	87.8	82.8	61.8	52.4

Table 7: DistilBERT, mean accuracy versus programming noise σ.

A.2 Accuracy versus SLC share (mean of 20 chips)

Noise	Placement	0%	5%	10%	20%	30%	50%	100%
3%	Magnitude	82.1	**89.7**	89.9	89.8	89.8	89.7	89.2
	Fisher		89.6	89.8	89.8	89.8	**89.8**	
	Gradient		88.7	**90.0**	**90.0**	**89.9**	89.6	
	Random		87.0	86.9	87.6	87.3	89.0	
5%	Magnitude	66.8	**88.0**	88.6	88.7	**88.9**	88.9	88.0
	Fisher		87.9	**88.6**	**88.8**	88.9	**88.9**	
	Gradient		84.2	88.2	88.4	88.4	88.1	
	Random		72.5	73.6	76.2	77.8	83.4	
7.5%	Magnitude	53.0	79.4	81.8	84.1	**85.5**	85.6	85.7
	Fisher		**80.0**	**82.2**	**84.2**	85.1	**85.8**	
	Gradient		67.5	79.0	81.2	82.4	83.2	
	Random		53.6	54.2	55.2	57.5	67.3	

Table 8: BERT-Base, mean accuracy by share of components in SLC (0% = all MLC, 100% = all SLC).

Noise	Placement	0%	5%	10%	20%	30%	50%	100%
3%	Magnitude	86.0	88.4	88.6	**88.8**	**88.9**	88.7	88.7
	Fisher		**88.5**	**88.7**	88.7	88.8	88.6	
	Gradient		88.4	88.5	88.6	88.8	**88.7**	
	Random		86.4	86.8	87.5	88.1	88.4	
5%	Magnitude	74.8	**87.4**	**88.0**	88.1	**88.8**	**88.4**	88.2
	Fisher		87.3	87.8	**88.4**	88.1	88.3	
	Gradient		87.3	87.3	87.7	88.1	87.8	
	Random		75.4	77.4	81.9	82.3	85.7	
7.5%	Magnitude	56.9	80.2	**84.0**	85.2	**86.6**	86.9	87.4
	Fisher		**80.8**	83.8	**85.4**	86.0	**87.0**	
	Gradient		78.0	80.7	83.1	84.8	86.0	
	Random		56.4	58.2	63.1	64.7	73.7	

Table 9: DistilBERT, mean accuracy by share of components in SLC (0% = all MLC, 100% = all SLC).

A.3 Accuracy versus SLC share (worst of 20 chips)

Noise	Placement	0%	5%	10%	20%	30%	50%	100%
3%	Magnitude	70.9	**88.9**	**89.2**	88.9	88.9	88.5	87.7
	Fisher		88.5	88.9	88.6	**89.1**	**88.8**	
	Gradient		87.5	89.2	**89.2**	88.9	88.5	
	Random		83.9	82.2	85.7	84.4	86.0	
5%	Magnitude	51.6	**86.4**	**87.6**	**88.0**	**88.1**	86.9	85.8
	Fisher		86.2	87.0	87.8	87.7	**87.4**	
	Gradient		77.3	86.5	86.9	86.5	85.4	
	Random		57.8	61.5	63.5	52.4	70.9	
7.5%	Magnitude	49.1	**73.4**	**76.8**	76.0	**82.1**	**83.0**	82.3
	Fisher		72.2	72.5	**78.4**	81.4	82.9	
	Gradient		52.3	72.8	75.8	70.0	76.6	
	Random		49.2	50.6	50.9	49.1	49.3	

Table 10: BERT-Base, worst-chip accuracy by share of components in SLC (0% = all MLC, 100% = all SLC).

Noise	Placement	0%	5%	10%	20%	30%	50%	100%
3%	Magnitude	83.6	**87.7**	**87.4**	87.3	88.0	**87.7**	87.5
	Fisher		87.7	87.4	**87.6**	87.8	87.5	
	Gradient		87.6	87.3	87.2	**88.1**	87.3	
	Random		84.3	84.1	85.9	87.3	86.5	
5%	Magnitude	62.3	85.6	**86.0**	85.7	**87.3**	**87.5**	86.8
	Fisher		85.8	86.0	**86.5**	87.0	86.8	
	Gradient		**85.9**	85.8	85.8	86.1	85.4	
	Random		67.3	67.4	77.8	63.2	77.3	
7.5%	Magnitude	50.1	72.5	**79.5**	82.8	83.4	85.0	85.9
	Fisher		**76.3**	79.5	**83.6**	**83.8**	**85.9**	
	Gradient		72.9	73.3	79.9	83.4	84.4	
	Random		50.8	50.7	52.1	50.9	54.5	

Table 11: DistilBERT, worst-chip accuracy by share of components in SLC (0% = all MLC, 100% = all SLC).

A.4 Stuck-at faults (20 chips)

Placement	0	0.1%	0.2%	0.5%	1%	2%	5%
mean							
All SLC	89.9	88.3	85.6	79.0	62.7	51.4	49.6
All MLC	90.0	58.6	52.6	50.8	51.2	49.6	49.6
Magnitude	89.7	89.6	**88.6**	**76.8**	**57.9**	**51.0**	50.0
Fisher	89.1	**89.7**	88.4	75.6	55.5	49.5	**50.3**
Gradient	89.8	89.4	87.5	71.0	52.8	50.2	49.5
Random	**90.0**	67.9	55.8	50.7	50.6	50.2	49.9
worst chip							
All SLC	89.9	83.0	71.3	57.5	50.9	49.1	41.3
All MLC	90.0	49.1	49.1	48.9	48.9	46.3	48.3
Magnitude	89.7	**89.1**	**86.2**	**66.2**	**50.8**	47.0	46.2
Fisher	89.1	88.1	86.1	57.7	49.1	44.8	**47.6**
Gradient	89.8	85.9	82.1	53.6	49.1	46.7	47.4
Random	**90.0**	50.8	49.1	47.4	48.9	**48.6**	46.2

Table 12: BERT-Base, accuracy versus share of stuck cells.

Placement	0	0.1%	0.2%	0.5%	1%	2%	5%
mean							
All SLC	89.0	88.2	87.8	85.2	74.6	58.5	51.5
All MLC	89.1	65.5	58.2	52.3	51.1	50.2	49.4
Magnitude	89.1	**88.3**	**87.4**	80.3	**62.1**	**51.4**	49.9
Fisher	89.0	88.3	87.2	**80.3**	62.1	51.3	50.4
Gradient	**89.8**	88.1	86.7	74.5	55.4	51.0	**50.5**
Random	88.6	76.9	63.3	51.3	51.0	49.4	50.1
worst chip							
All SLC	89.0	86.9	85.3	79.8	58.1	50.9	46.4
All MLC	89.1	50.9	49.3	49.1	49.1	45.6	45.4
Magnitude	89.1	85.6	83.0	50.9	**51.0**	44.4	46.4
Fisher	89.0	86.4	**85.2**	50.9	49.1	44.4	**47.7**
Gradient	**89.8**	**86.7**	83.0	**51.5**	50.9	**45.8**	46.8
Random	88.6	52.4	50.0	47.5	46.8	44.6	46.7

Table 13: DistilBERT, accuracy versus share of stuck cells.

A.5 Read noise (5 chips)

Placement	0	0.5%	1%	2%	3%	5%
mean						
All SLC	89.7	89.8	89.8	89.0	87.2	80.0
All MLC	90.4	89.4	87.7	77.3	63.8	53.0
Magnitude	**90.6**	90.1	89.9	**90.1**	**89.3**	82.5
Fisher	89.6	90.0	89.9	89.7	89.0	**82.7**
Gradient	90.3	**90.5**	**90.3**	89.8	88.3	76.6
worst chip						
All SLC	89.7	89.1	89.4	88.6	86.0	79.2
All MLC	90.4	89.1	86.9	76.0	62.4	52.2
Magnitude	**90.6**	89.7	**89.3**	88.5	**89.0**	81.1
Fisher	89.6	89.7	89.3	**89.1**	88.6	**81.2**
Gradient	90.3	**89.8**	89.2	89.0	87.7	75.7

Table 14: BERT-Base, read noise / $G_{\max}$.

Placement	0	0.5%	1%	2%	3%	5%
mean						
All SLC	88.5	89.0	88.2	88.3	87.5	84.5
All MLC	89.0	88.6	87.9	83.8	74.5	57.7
Magnitude	88.8	**88.9**	**88.7**	**88.4**	**87.7**	**83.3**
Fisher	**89.1**	88.8	88.7	88.2	87.5	83.2
Gradient	88.6	88.9	88.5	88.1	86.8	80.5
worst chip						
All SLC	88.5	88.0	87.8	88.0	87.0	84.1
All MLC	89.0	88.1	87.4	82.2	73.6	57.0
Magnitude	88.8	**88.5**	87.5	**88.0**	86.1	**82.5**
Fisher	**89.1**	88.3	87.7	87.6	**87.0**	82.1
Gradient	88.6	88.4	**88.0**	87.3	85.9	79.4

Table 15: DistilBERT, read noise / $G_{\max}$.

A.6 Level-dependent MLC noise at $\sigma = 5\%$ (10 chips)

Placement	1/1/1/1	1/1.5/1.5/1	1/2/2/1
mean			
All SLC	88.2	88.2	88.2
All MLC	67.3	64.7	62.5
Magnitude	88.9	88.7	88.4
Fisher	**89.0**	**88.9**	**88.4**
Gradient	88.3	88.1	87.8
worst chip			
All SLC	86.2	86.2	86.2
All MLC	57.7	54.7	53.2
Magnitude	87.7	87.7	**87.4**
Fisher	**88.2**	**88.2**	86.9
Gradient	87.3	86.6	86.2

Table 16: BERT-Base, noise multiplier for MLC levels 0/1/2/3.

Placement	1/1/1/1	1/1.5/1.5/1	1/2/2/1
mean			
All SLC	88.4	88.4	88.4
All MLC	75.7	73.5	70.3
Magnitude	**88.4**	**88.3**	**88.0**
Fisher	88.1	88.0	87.9
Gradient	88.0	87.5	87.1
worst chip			
All SLC	87.4	87.4	87.4
All MLC	61.6	56.9	54.1
Magnitude	**87.8**	**87.2**	**86.6**
Fisher	86.4	86.7	85.8
Gradient	87.3	86.4	85.6

Table 17: DistilBERT, noise multiplier for MLC levels 0/1/2/3.

A.7 1% stuck cells by fault pattern (10 chips)

Placement	uniform	cluster	line
mean			
All SLC	61.3	66.7	57.0
All MLC	50.7	50.8	50.4
Magnitude	**56.4**	**54.0**	50.4
Fisher	55.6	53.1	**52.2**
Gradient	52.6	51.5	51.0
worst chip			
All SLC	50.9	49.1	50.9
All MLC	49.1	49.1	49.1
Magnitude	**51.0**	49.1	46.9
Fisher	49.3	**50.9**	**49.8**
Gradient	49.9	49.2	49.1

Table 18: BERT-Base, fault pattern.

Placement	uniform	cluster	line
mean			
All SLC	74.2	80.3	71.2
All MLC	51.2	51.2	50.8
Magnitude	**62.5**	56.1	54.7
Fisher	60.0	**57.3**	**56.3**
Gradient	55.0	56.4	52.2
worst chip			
All SLC	58.7	73.7	50.9
All MLC	48.5	48.9	46.8
Magnitude	**52.2**	**51.6**	**50.9**
Fisher	49.1	50.9	50.9
Gradient	50.9	50.9	48.6

Table 19: DistilBERT, fault pattern.

A.8 IR drop (deterministic)

Placement	0	1×10^{-4}	3×10^{-4}	1×10^{-3}	3×10^{-3}
All SLC	89.7	90.3	88.9	87.0	58.5
All MLC	90.4	89.3	89.8	90.6	79.1
Magnitude	**90.6**	90.1	90.4	88.1	**76.8**
Fisher	89.6	**91.1**	**91.1**	88.3	72.8
Gradient	90.3	90.5	90.5	**88.6**	76.4

Table 20: BERT-Base, $r = R_{\text{wire}}/R_{\text{on}}$ per segment.

Placement	0	1×10^{-4}	3×10^{-4}	1×10^{-3}	3×10^{-3}
All SLC	88.5	89.9	89.4	85.3	70.1
All MLC	89.0	88.6	89.1	89.9	77.9
Magnitude	88.8	89.2	**89.7**	86.6	73.4
Fisher	**89.1**	89.2	89.3	86.4	72.6
Gradient	88.6	**89.6**	89.6	**86.9**	**74.7**

Table 21: DistilBERT, $r = R_{\text{wire}}/R_{\text{on}}$ per segment.

A.9 Fixed ADC with bit-serial inputs, no noise (deterministic)

Placement	ideal	8	7	6	5	4	3
All SLC	89.7	89.9	90.0	89.3	51.1	61.1	51.1
All MLC	90.4	89.6	89.2	83.0	51.1	60.3	50.9
Magnitude	**90.6**	**90.4**	89.3	90.0	51.1	59.9	52.6
Fisher	89.6	90.1	89.7	**90.5**	51.6	57.0	**52.9**
Gradient	90.3	90.1	**90.1**	89.6	**53.0**	**63.2**	51.9

Table 22: BERT-Base, ADC bits.

Placement	ideal	8	7	6	5	4	3
All SLC	88.5	89.2	88.4	88.0	71.0	52.1	59.1
All MLC	89.0	88.3	87.8	81.7	56.5	51.9	56.8
Magnitude	88.8	**88.6**	**89.0**	**89.7**	70.6	51.4	**62.4**
Fisher	**89.1**	88.0	88.2	88.8	70.9	**51.7**	60.3
Gradient	88.6	88.4	88.5	88.4	**77.3**	51.5	60.1

Table 23: DistilBERT, ADC bits.

A.10 Fixed ADC with bit-serial inputs at $\sigma = 5\%$ (DistilBERT, 3 chips)

Placement	ideal	8	6	4
mean				
All SLC	88.1	88.3	88.7	52.6
All MLC	77.8	76.8	56.0	53.6
Magnitude	**88.7**	**89.1**	**88.5**	51.5
Fisher	87.8	88.5	88.2	**52.3**
Gradient	87.8	87.6	87.2	51.8
worst chip				
All SLC	87.4	87.8	87.5	52.2
All MLC	69.7	68.2	51.4	50.9
Magnitude	**87.8**	**88.1**	**88.3**	51.3
Fisher	86.4	87.2	87.7	**51.9**
Gradient	87.3	86.9	87.0	51.4

Table 24: DistilBERT, ADC bits.